FIRST TERM CONTINUOUS ASSESSMENT 1 2026/2027

SUBJECT: MATHEMATICS	**SECTION(S):**A	**12TH GRADE**

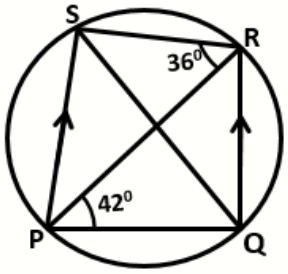
TIME ALLOWED: 20MINUTES

SECTION A: OBJECTIVES

1. If $\sin\theta = -\frac{1}{2}$, find all the values of θ between 0^0 and 360^0.
   A. 210^0, 330^0
   B. 210^0, 300^0
   C. 300^0, 360^0
   D. 120^0, 240^0

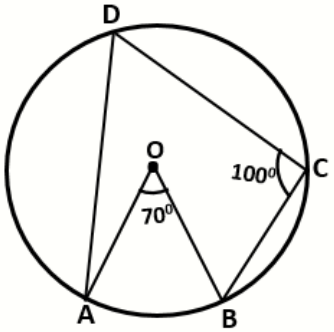
2. Abidjan is 4^0 West of Accra and on the same circle of latitude. If the radius of this circle of latitude is 6370km, how far is Abidjan West of Accra, correct to the nearest kilometer? (Take $\pi = \frac{22}{7}$)
   A. 222km
   B. 445km
   C. 890km
   D. 5005km

3. In the diagram, PS//QR. If $\angle QPR = 42^0$ and $\angle PRS = 36^0$, find $\angle PSQ$.
   A. 51^0
   B. 52^0
   C. 53^0
   D. 54^0



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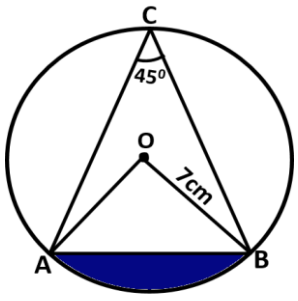
4. In the diagram, O is the centre of the circle. If $\angle AOB = 70^0$ and $\angle BCD = 100^0$, find $\angle DAO$.
   A. 35^0
   B. 30^0
   C. 25^0
   D. 20^0



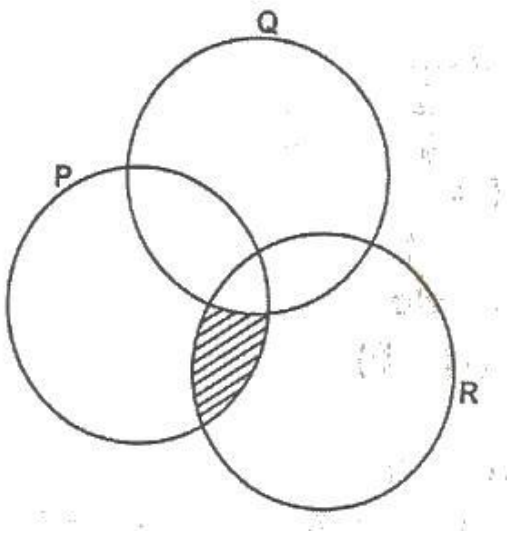
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5. The implication x → y is equivalent to
   A. $\sim y \rightarrow \sim x$
   B. $y \rightarrow \sim x$
   C. $\sim x \rightarrow \sim y$
   D. $y \rightarrow x$

6. ABC is a circle centre O and radius 7cm. Find the area of the shaded region. (Take $\pi = \frac{22}{7}$).
   A. $13cm^2$
   B. $14cm^2$
   C. $15cm^2$
   D. $16cm^2$



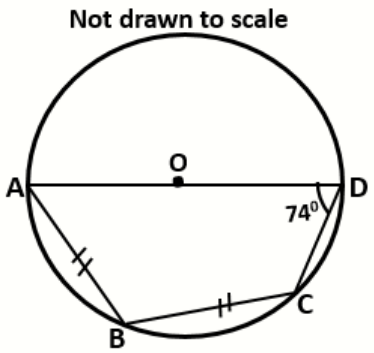
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7. In the Venn diagram, the shaded region is
   A. (P ∩ Q) ∪ R
   B. (P ∩ Q) ∩ R
   C. (P ∩ Q^1) ∩ R
   D. (P ∩ Q^1) ∪ R

8. In a circle of radius 8cm, calculate, correct **to 3 significant figures,** the length of a chord which is 5cm from the centre of the circle.
   A. 12.4cm
   B. 12.5cm
   C. 13.0cm
   D. 13.5cm

9. In the diagram below, AB = BC and AOD is a diameter of the circle centre O. If $\angle ADC = 74^0$, find $\angle BAD$.



   A. 53^0
   B. 54^0
   C. 56^0
   D. 57^0

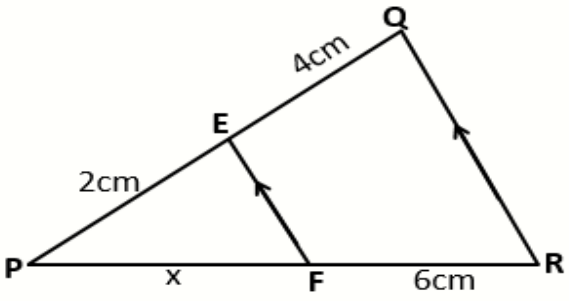
10. The ages of Tunde and Ola are in the ratio

    1 :2. If the ratio of Ola's age to Muas's age is

    4 :5, what is the ratio of Tunde's age to Muas's age?

   A. 1: 4
   B. 1 : 5
   C. 2 : 5
   D. 5 : 2

11. In the diagram, EF//QR, |PE| = 2cm, |EQ| = 4cm and |FR| = 6cm. Find x.

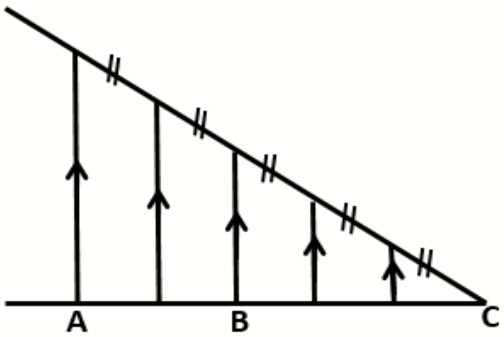


   A. 5cm
   B. 4cm
   C. 3cm
   D. 2cm

12. Given that t = 2^{-x} , find 2^{x+1} in terms of t.
   A. $\frac{2}{t}$
   B. $\frac{t}{2}$
   C. $\frac{1}{2t}$
   D. 2t

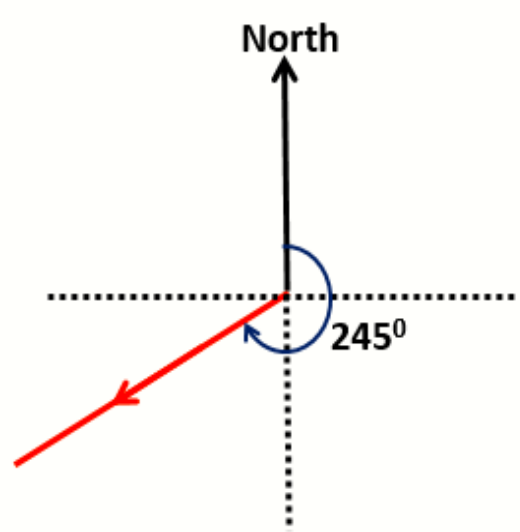
13. A bag contains blue, yellow and green identical balls. The probability of picking a blue ball at random from the bag is $\frac{2}{5}$ and the probability of picking a yellow ball at random is $\frac{3}{10}$. If there are 40 balls in the bag, how many of them are green?
   A. 3
   B. 7
   C. 9
   D. 12

14. In the diagram below, find the ratio $\frac{BC}{AC}$.



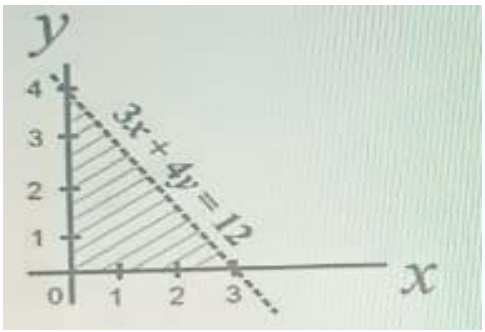
   A. $\frac{2}{5}$
   B. $\frac{3}{5}$
   C. $\frac{2}{3}$
   D. $\frac{5}{3}$

15. What is the cardinal direction shown by the diagram below?



A. $N65^0W$
B. $S25^0E$
C. $N25^0E$
D. $S65^0W$

16. The graph below represents which inequalities?



A. $3x + 4y \geq 12, x > 0, y > 0$
B. $3x + 4y \leq 12, x \geq 0, y \geq 0$
C. $3x + 4y < 12, x \geq 0, y \geq 0$
D. $3x + 4y < 12, x \leq 0, y \geq 0$

17. A ladder 12m long is placed against a vertical wall. The foot of the ladder is 5m from the wall. Find the angle of elevation of the ladder to the wall, correct to the nearest 0.1^0.
A. 24.6^0
B. 55.3^0
C. 65.4^0
D. 65.7^0

18. Two buses start from the same station at 9:00a.m and travel in opposite directions along the same straight road. The first bus travels at a speed of 72km/h and the second at 48km/h. At what time will they be 240km apart?
A. 1:00pm
B. 12:00noon
C. 11:00 am
D. 10:00am

19. The sum of the age of 14 learners in a class is 168. A learner was added to the class and the mean age changed to 12.2. Calculate the age of the new learner.
A. 13
B. 15
C. 14
D. 16

20. In an athletic competition, the probability that an athlete wins a 100m race is $\frac{1}{8}$ and the probability that he wins in high jump is $\frac{1}{4}$. What is the probability that he wins only one of the events?
A. $\frac{3}{32}$
B. $\frac{3}{16}$
C. $\frac{7}{32}$
D. $\frac{5}{16}$